

Store Spatial Sales Analytics

Work Overview · SQL Server → Microsoft Fabric → Direct Lake → Power BI

Overall status: **WORK IN PROGRESS**

Project Objective

Build an end-to-end spatial sales reporting solution that begins with a local SQL Server database, ingests data into Microsoft Fabric, adds geographic and date dimensions, creates a Direct Lake semantic model, and produces an interactive Power BI dashboard for analysis by state, category and time.

Progress Summary

Stage	Status	Current Position
SQL Server / StoreDB	Completed	StoreDB created, normalized and populated with the sales dataset.
Gateway connection	Completed	Local SQL Server connected to Microsoft Fabric.
Fabric ingestion	Completed / refining	PL_Load_StoreDB created and Copy Data activities configured.
Geographic data	Completed	StateGeography created with state codes, latitude and longitude.
Date dimension	Completed	DateDim created and loaded for time-intelligence reporting.
Semantic model	Completed	Direct Lake model created with table relationships and DAX measures.
Power BI dashboard	In progress	Slicers, KPI areas, map, trend and category visuals are being built and refined.
Spatial mapping	In progress	State mapping is present; custom boundary/GeoJSON and final spatial behaviour still being refined.
Testing / documentation	In progress	Visual interactions, DAX results, final screenshots and GitHub documentation still being completed.

Work Currently in Progress

- Completing and formatting the KPI cards for Total Sales, Total Profit, Profit Margin % and Total Orders.
- Refining state-level spatial mapping and the custom Shape Map / GeoJSON requirement.

- Improving the report heading, logo, spacing, colours and overall visual consistency.
- Adding/refining the state performance matrix and dynamic High / Medium / Low territory segmentation.
- Testing Year, Region, State, Category, Segment and Ship Mode slicers.
- Validating Sales YTD, Previous Year Sales and YoY Growth % calculations.
- Testing cross-filtering between the map, trend visual, category analysis, cards and slicers.
- Preparing final screenshots and GitHub documentation once the dashboard is complete.

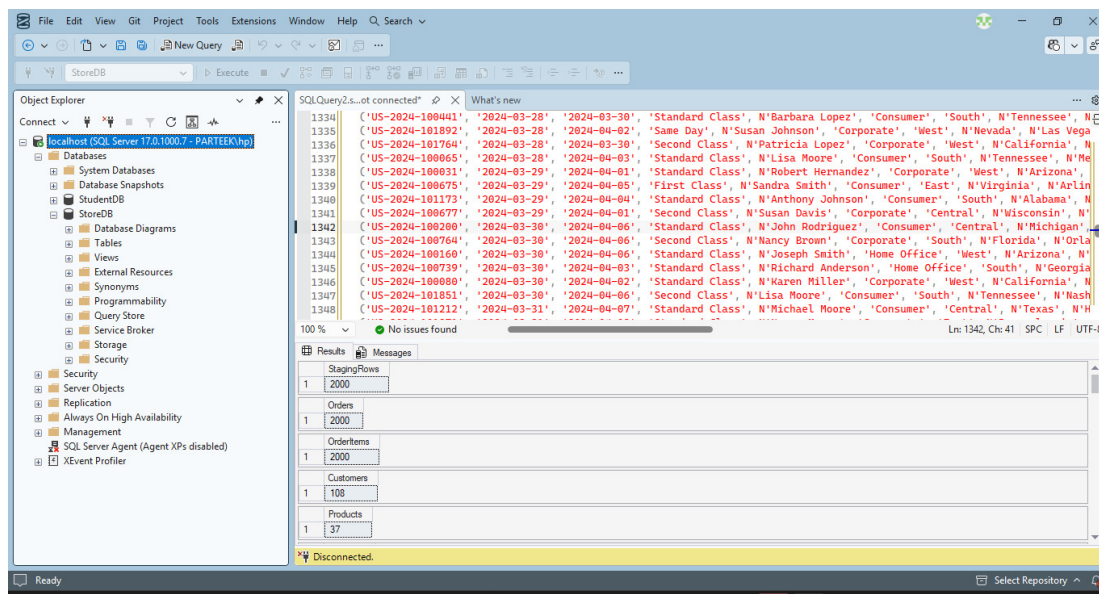
Planned Final Dashboard

The final dashboard will contain a title/logo area, six slicers, KPI cards, a state-level spatial map, time-intelligence trend analysis, category analysis, a state performance matrix, dynamic territory segmentation and fully tested interactive filtering.

Project Evidence

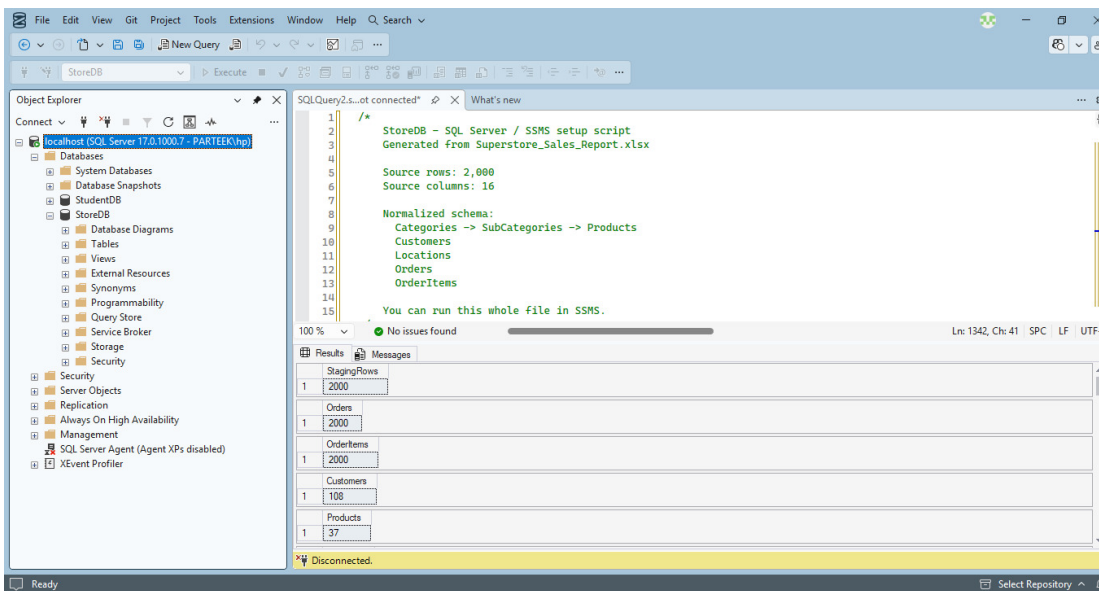
Screenshots documenting progress across the SQL Server, Microsoft Fabric and Power BI stages of the build.

Screenshot 1



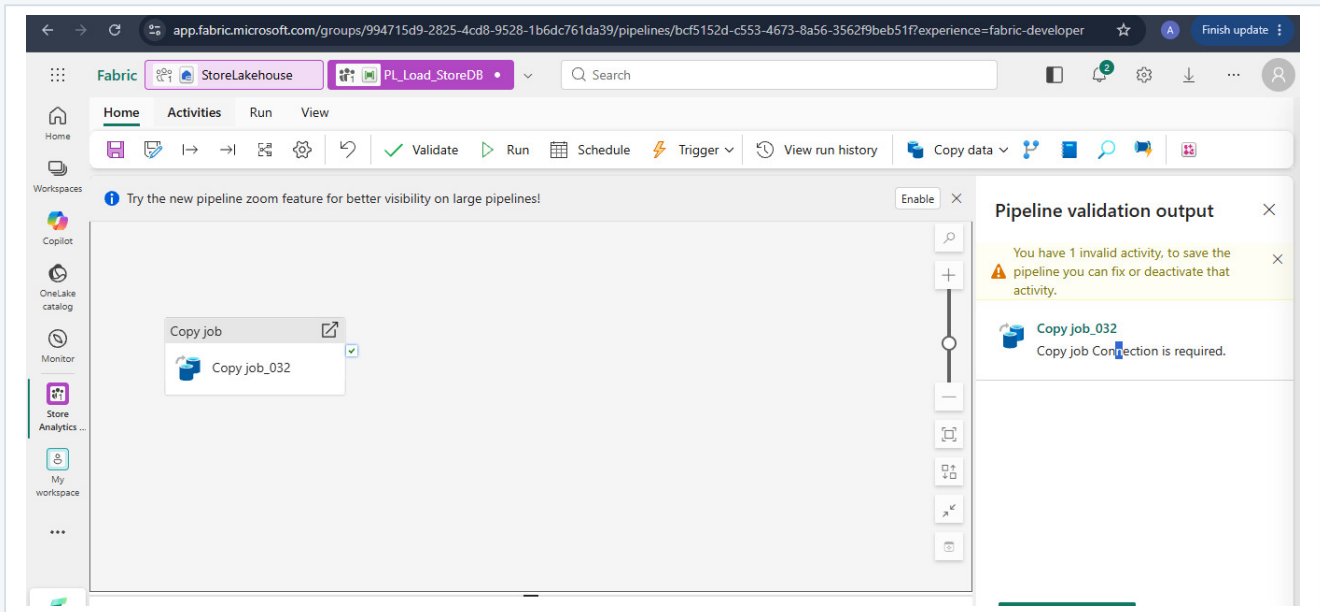
SQL Server database preparation in SSMS

Screenshot 2



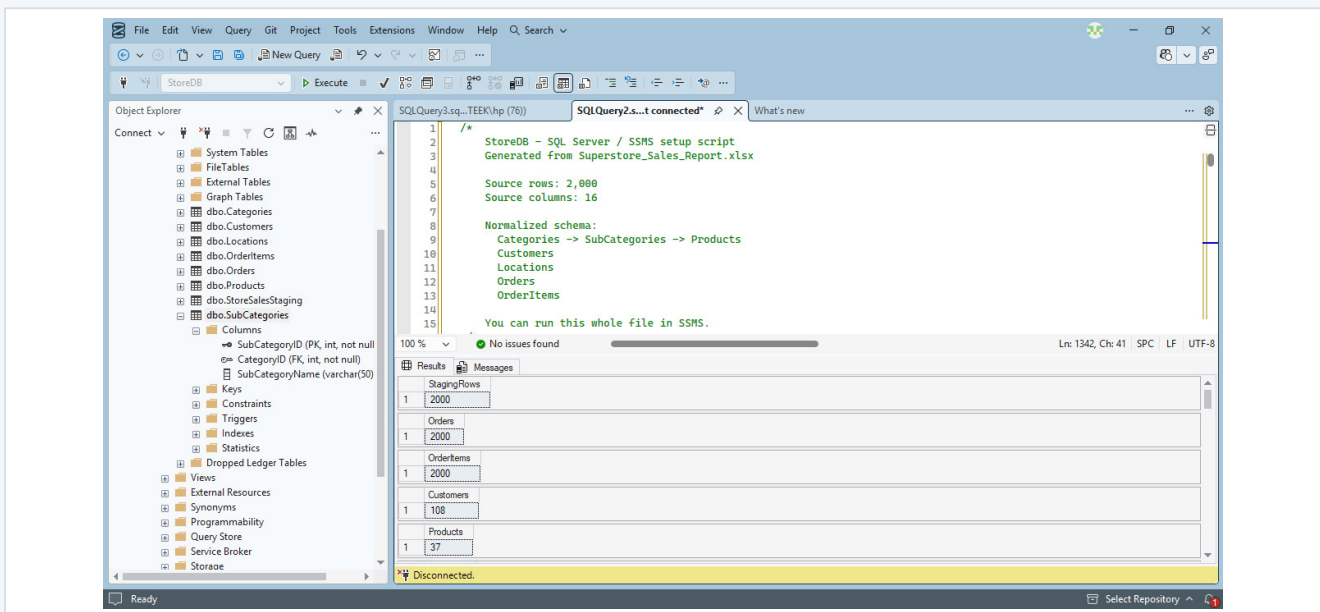
StoreDB created and populated from the Superstore dataset

Screenshot 3



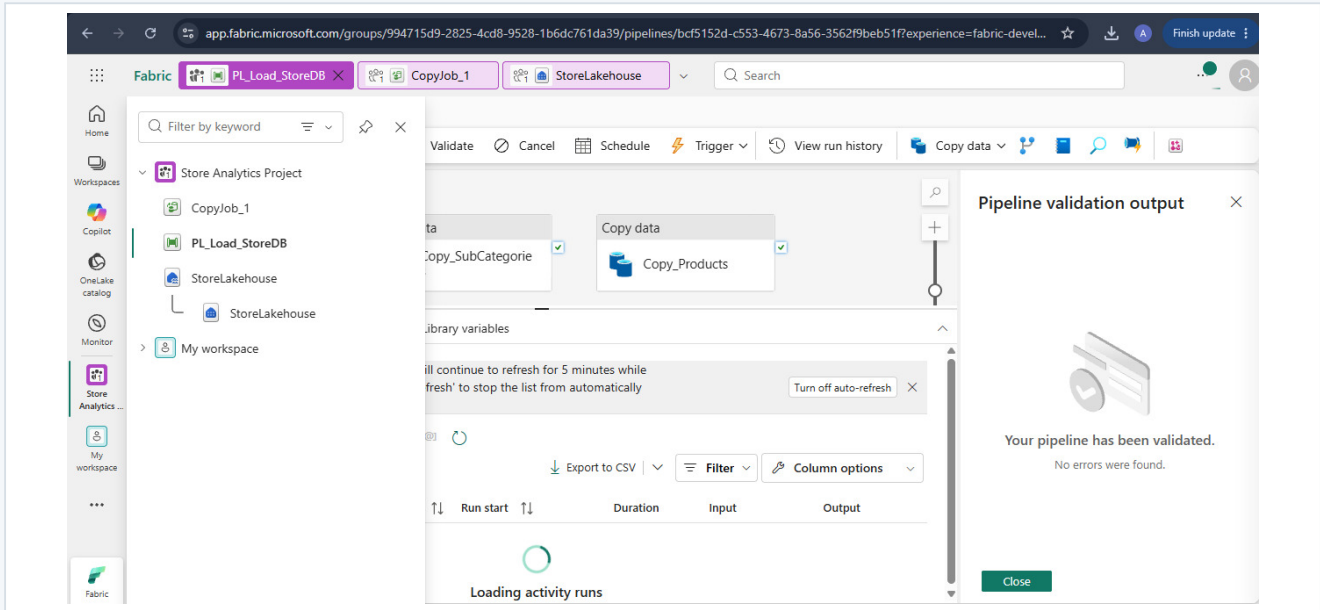
Early Fabric pipeline troubleshooting: Copy Job validation error

Screenshot 4



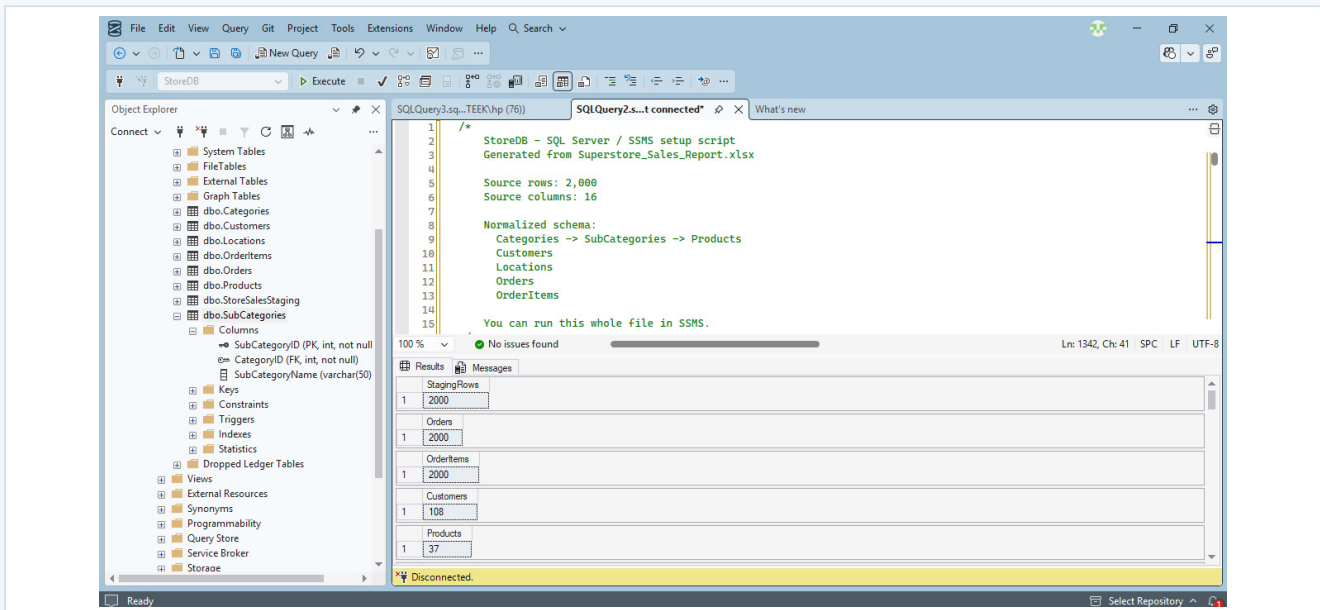
Normalized StoreDB tables visible in SQL Server Management Studio

Screenshot 5



Fabric pipeline after moving to Copy Data activities

Screenshot 6



Fabric workspace and StoreLakehouse pipeline configuration

Screenshot 7

The screenshot shows a SQL Server Enterprise Manager window with a query executed successfully. The query is a SELECT statement that returns the StateName, StateCode, Latitude, and Longitude for various US states. The results are displayed in a table with 12 rows and 4 columns.

StateName	StateCode	Latitude	Longitude
Alabama	AL	32.806671	-86.791130
Arizona	AZ	33.729759	-111.431221
California	CA	36.116203	-119.681564
Florida	FL	27.766279	-81.686783
Georgia	GA	33.040619	-83.643074
Illinois	IL	40.349457	-88.986137
Massach...	MA	42.230171	-71.530105
Michigan	MI	43.266618	-84.536095
Nevada	NV	38.313515	-117.055374
New Jersey	NJ	40.289904	-74.521011
New York	NY	42.165726	-74.940511
North Car...	NC	35.630066	-79.806419

StateGeography table with state codes, latitude and longitude

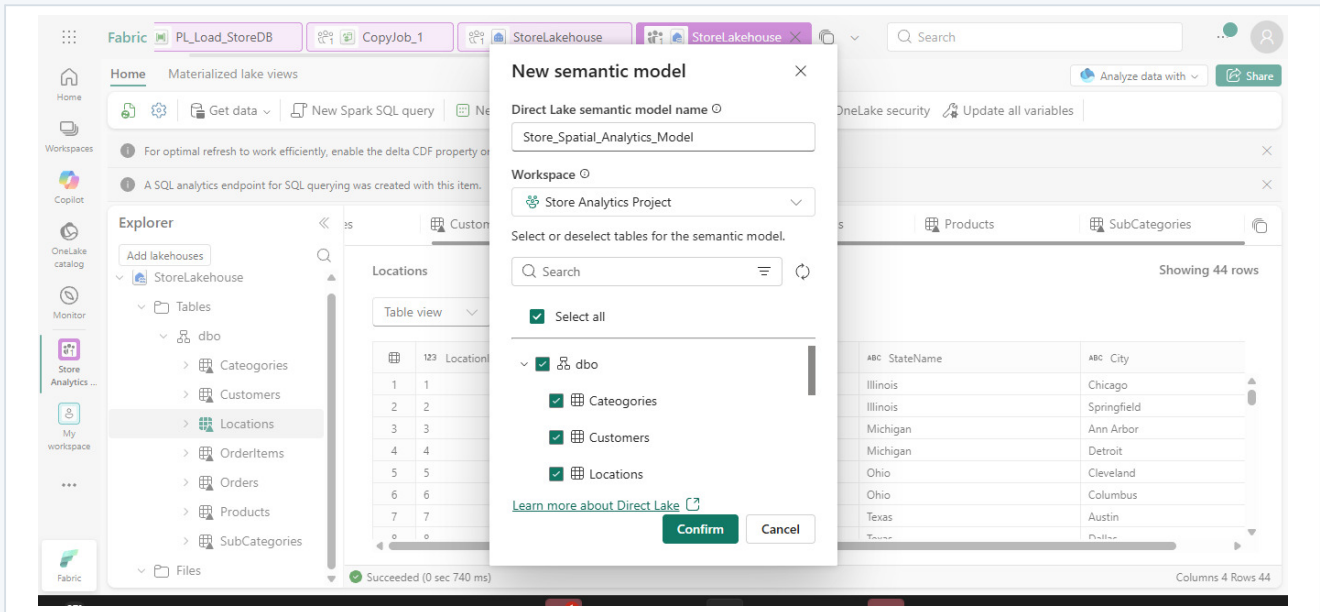
Screenshot 8

The screenshot shows the Microsoft Fabric portal interface. The top navigation bar includes tabs for Home, Activities, Run, and View. The main area displays a pipeline validation output for a pipeline named 'CopyJob_1'. The pipeline consists of several steps, including 'Copy_Orders', 'Copy_Categories', 'Copy_DateDim', 'Copy_Products', 'Copy_OrderItems', 'Copy_Locations', and 'Copy_SubCategories'. The validation output shows that the pipeline has been validated successfully, with no errors found.

Parameters	Variables	Settings	Output	Library variables
Copy_Orders		In progress	9/18/2026, 11:06:38 PM	48s
Copy_Categories		Succeeded	9/18/2026, 11:06:38 PM	45s
Copy_DateDim		In progress	9/18/2026, 11:06:38 PM	48s
Copy_Products		In progress	9/18/2026, 11:06:38 PM	48s
Copy_OrderItems		In progress	9/18/2026, 11:06:38 PM	48s
Copy_Locations		In progress	9/18/2026, 11:06:38 PM	48s
Copy_SubCategories		In progress	9/18/2026, 11:06:38 PM	48s

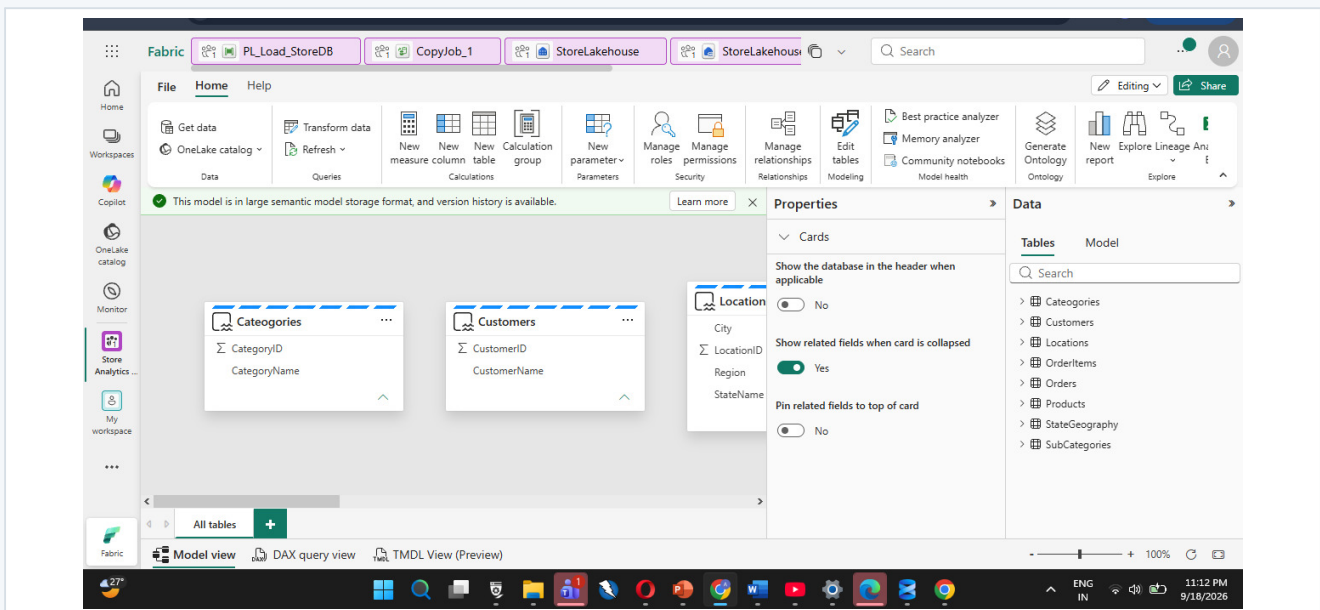
Expanded Fabric pipeline including DateDim and StoreDB tables

Screenshot 9



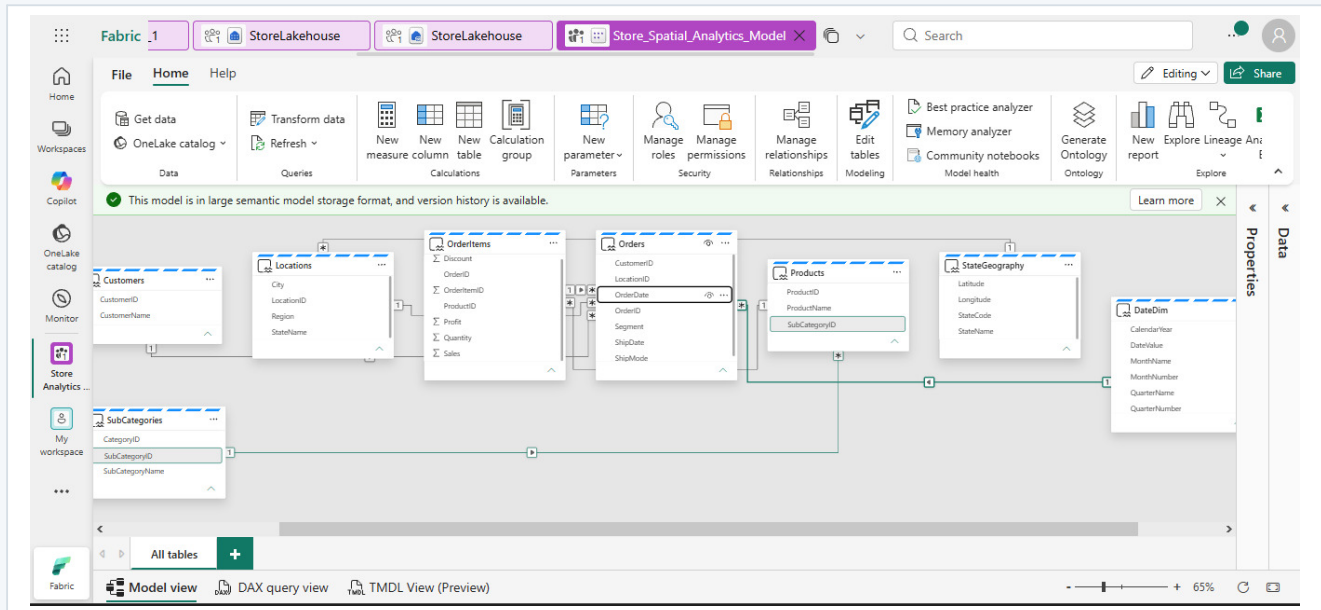
Creating the Direct Lake semantic model from StoreLakehouse

Screenshot 10



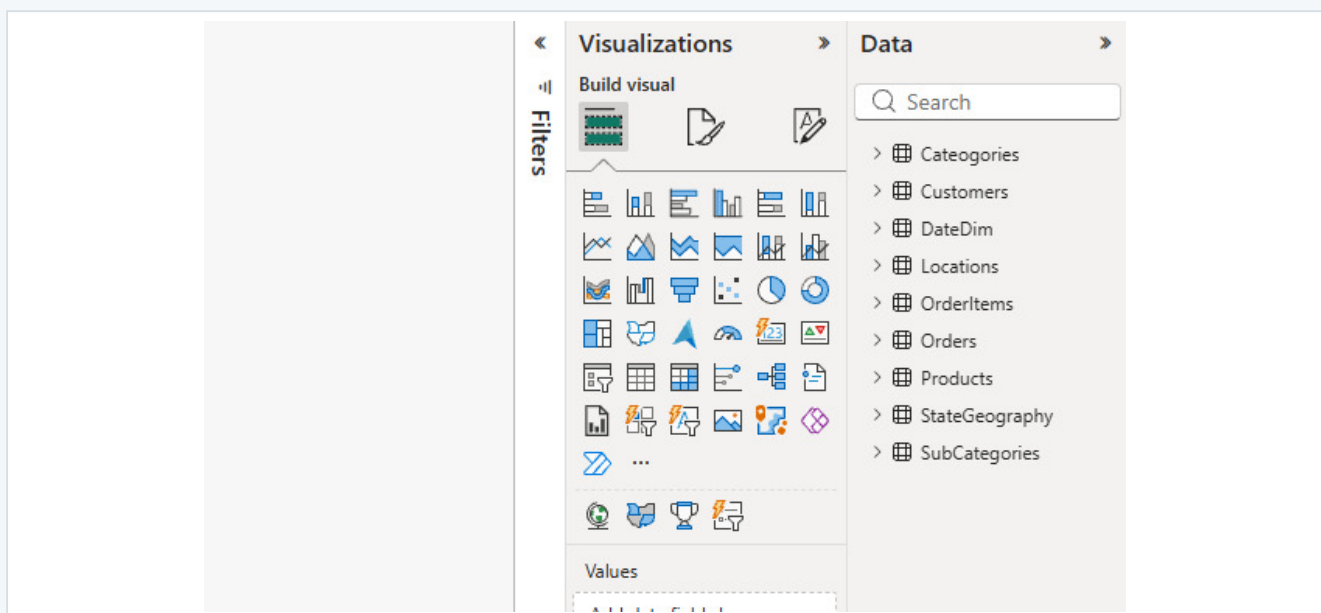
Initial semantic model view in Microsoft Fabric

Screenshot 11



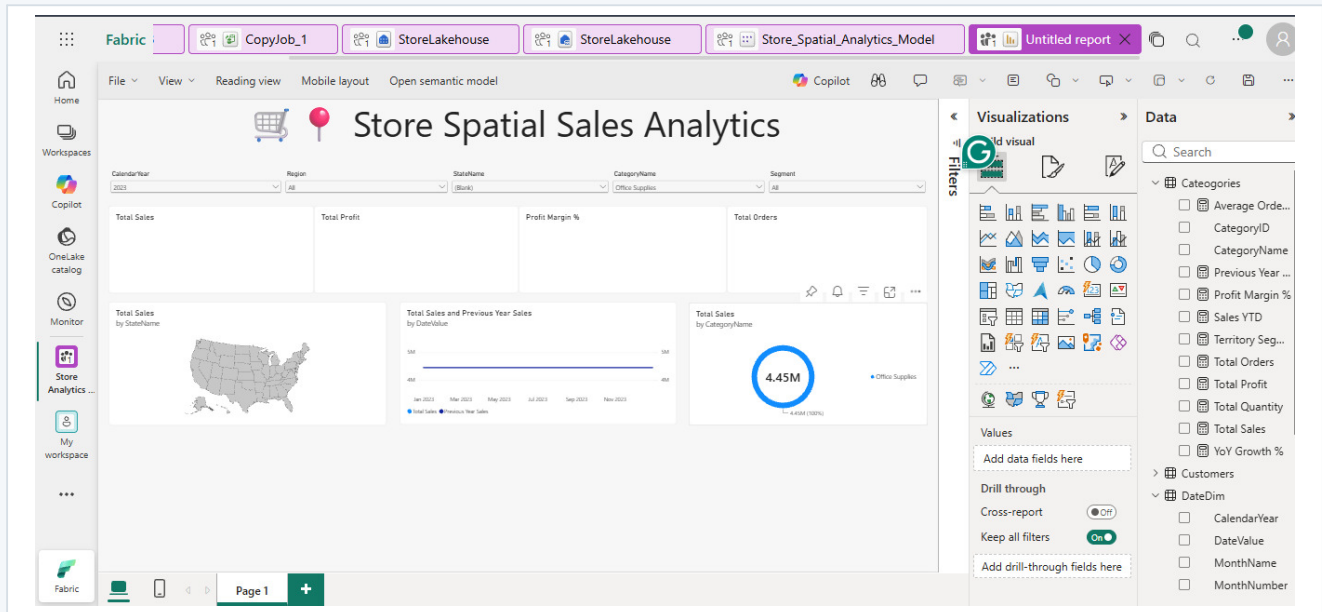
Semantic model relationships including StateGeography and DateDim

Screenshot 12



Power BI visual selection while building the report

Screenshot 13



Current Store Spatial Sales Analytics report — work in progress